Report

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Summary

Tasks Status

Task	Status	Link
Criar toy model		Notebook
↪ Gerar dados sintéticos (correlacionados + ruído opcional)	✓	
↪ Ajustar GP (kernel, hiperparâmetros)	✓	
↪ Comparar: resíduos, log-likelihood	✓	
↪ Plotar a previsão	✓	

Tasks Status

Task	Status	
Analisar toy model		
↪ Sensibilidade à escolha do kernel	○	
↪ Otimização do log marginal	○	
↪ Predição	○	
↪ Resíduos	○	
↪ Comparação: modelo correto vs kernel errado	○	

Tasks Status

Task	Status	Link
Escrever projeto		Projeto
↪ Definir escopo e objetivos	✓	
↪ Esboçar introdução (motivação + observáveis)	✓	
↪ Listar método/dados (GP, Inferencia Bayesiana)	○	
↪ Listar objetivos	◑	
↪ Cronograma	✓	
↪ Selecionar bibliografia	✓	

Recent studies

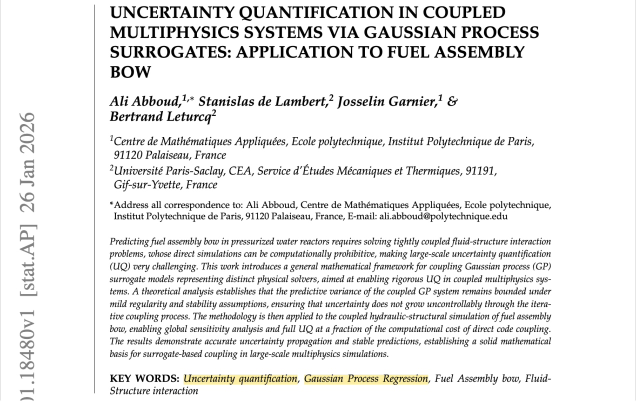
Article	Status	Link
The BRAHMS experiment at RHIC	✓	Article
The ALICE experiment at the CERN LHC	◑	Article
The ALICE Time-Of-Flight detector at the LHC	◑	Article
The ALICE TPC: a large 3-dimensional tracking device with fast readout for ultra-high multiplicity events	◑	Article

Recent studies

Toolbox	**Status**	**Link**
2. Gaussian Processes	◑	Lecture Notes

Course	**Status**	**Link**
Statistics For Applications	✓	MIT OCW 18.650

Selected Paper



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UNCERTAINTY QUANTIFICATION IN COUPLED MULTIPHYSICS SYSTEMS VIA GAUSSIAN PROCESS SURROGATES: APPLICATION TO FUEL ASSEMBLY BOW

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Predicting fuel assembly bow in pressurized water reactors requires solving tightly coupled fluid-structure interaction problems, whose direct simulations can be computationally prohibitive, making large-scale uncertainty quantification (UQ) very challenging. This work introduces a general mathematical framework for coupling Gaussian process (GP) surrogate models representing distinct physical solvers, aimed at enabling rigorous UQ in coupled multiphysics systems. A theoretical analysis establishes that the predictive variance of the coupled GP system remains bounded under mild regularity and stability assumptions, ensuring that uncertainty does not grow uncontrollably through the iterative coupling process. The methodology is then applied to the coupled hydraulic-structural simulation of fuel assembly bow, enabling global sensitivity analysis and full UQ at a fraction of the computational cost of direct code coupling. The results demonstrate accurate uncertainty propagation and stable predictions, establishing a solid mathematical basis for surrogate-based coupling in large-scale multiphysics simulations.

KEY WORDS: *Uncertainty quantification, Gaussian Process Regression, Fuel Assembly bow, Fluid-Structure interaction*

Figure: This work analyses a Gaussian Process surrogate framework for uncertainty quantification in coupled multiphysics systems.